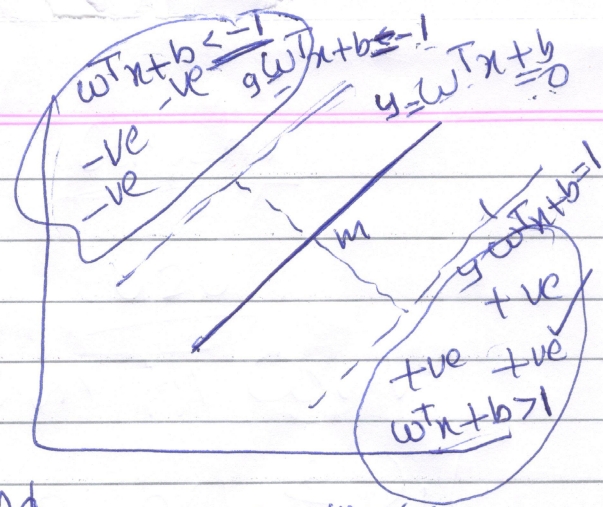


SVM =



margin maximizing
classification



$$\text{margin} = \frac{2}{\|w\|}$$

=

$$y_i = w^T x + b$$

$$y_i = (w^T x + b)$$

Find
weights = $\frac{w_1}{2}, \frac{w_2}{3}, \frac{w_3}{4}$

$$\|w\| = \sqrt{w_1^2 + w_2^2 + w_3^2}$$

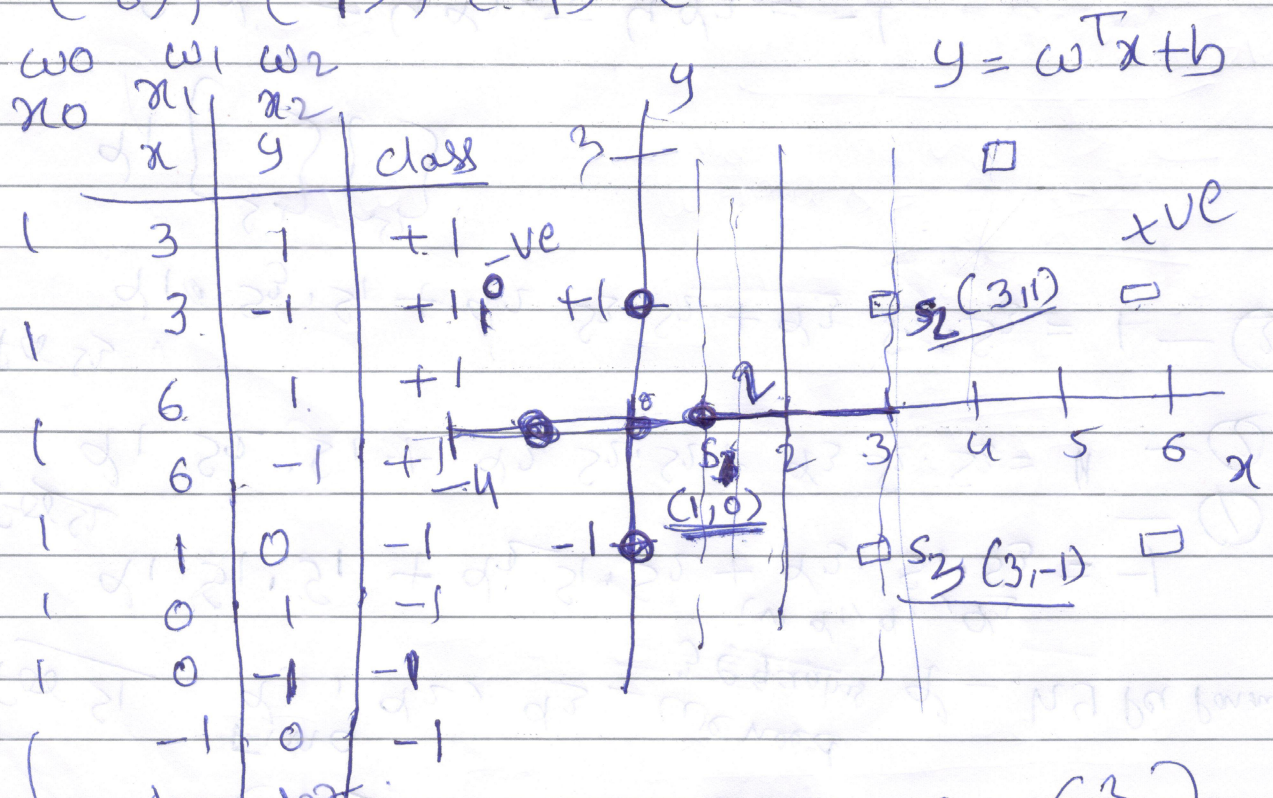
$$= \frac{2}{\sqrt{2^2 + 3^2 + 4^2}}$$

$$y = w^T x + b$$

$$y = \underline{w_1 x_1 + w_2 x_2 + w_3 x_3 + b}$$

$\begin{pmatrix} 3 \\ 1 \end{pmatrix}, \begin{pmatrix} 3 \\ -1 \end{pmatrix}, \begin{pmatrix} 6 \\ 1 \end{pmatrix}, \begin{pmatrix} 6 \\ -1 \end{pmatrix}$ - +ve class

$\begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ -1 \end{pmatrix}, \begin{pmatrix} -1 \\ 0 \end{pmatrix}$ - -ve class



3 support vectors
 $S_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ $S_2 = \begin{pmatrix} 3 \\ 1 \end{pmatrix}$ $S_3 = \begin{pmatrix} 3 \\ -1 \end{pmatrix}$

Find weights

$$w = \sum d_i S_i$$

we know S_1, S_2, S_3
 we have to find d_i i.e. d_1, d_2, d_3

Augmentation — Support vectors $\begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$

$$S_1 = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}$$

$$S_2 = \begin{pmatrix} 3 \\ 1 \\ 1 \end{pmatrix}$$

$$S_3 = \begin{pmatrix} 3 \\ -1 \\ 1 \end{pmatrix}$$

For S_1 Find d_1, d_2, d_3 — we need 3 equations of — by Per Param —

$$d_1 S_1 \cdot S_1 + d_2 S_1 \cdot S_2 + d_3 S_1 \cdot S_3 = -1$$

For S_2

$$d_1 S_2 \cdot S_1 + d_2 S_2 \cdot S_2 + d_3 S_2 \cdot S_3 = 1 \quad \text{--- (1)}$$

For S_3

$$d_1 S_3 \cdot S_1 + d_2 S_3 \cdot S_2 + d_3 S_3 \cdot S_3 = 1 \quad \text{--- (2)}$$

$$d_1 \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$$

$$2d_1 + 4d_2 + 4d_3 = -1$$

$$4d_1 + 11d_2 + 9d_3 = 1$$

$$4d_1 + 9d_3 + 11d_3 = 1$$

$$d_1 = -3.5$$

$$d_2 = 0.75$$

$$d_3 = 0.75$$

$$w = \sum d_i \cdot s_i$$

$$= -3.5 \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} + 0.75 \begin{bmatrix} 3 \\ 1 \\ 1 \end{bmatrix} + 0.75 \begin{bmatrix} 3 \\ -1 \\ 1 \end{bmatrix}$$

$$\begin{pmatrix} w_1 \\ w_2 \\ w_0 \end{pmatrix} = \begin{pmatrix} -3.5 \\ 0 \\ -3.5 \end{pmatrix} + \begin{pmatrix} 2.25 \\ 0.75 \\ 0.75 \end{pmatrix} + \begin{pmatrix} 2.25 \\ -0.75 \\ 0.75 \end{pmatrix}$$

$$\begin{pmatrix} w_1 \\ w_2 \\ b \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ -2 \end{pmatrix}$$

$$w_1 = 1$$

$$w_2 = 0$$

$$b = -2$$

hyperplane

eqn $y = w^T x + b$

$$= w_1 x_1 + w_2 x_2 + b$$

$$\boxed{y_1 = x_1 - 2}$$

Predict future data - to check if it is +ve class
-ve class

$$\text{Maximum margin} = \frac{2}{\|w\|}$$

$$= \frac{2}{\sqrt{1+0+2}} = \frac{2}{1} = 2$$

① $(2, 3)$
↓
adding +1
+ve

$$y_1 = x_1 - 2$$
$$= 4 - 2 = 2$$

$$y_1 = 2 > 1 - \text{+ve}$$
$$< -1$$

② $(-4, 1)$
↓
-1
-ve

$$y_1 = x_1 - 2$$

$$= -4 - 2 = -6 < -1$$

-ve